Do-it-yourself and benchtop DNA synthesis: a biosecurity assessment and a route-attribution capability

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Abstract

Commercial DNA providers can screen orders against databases of hazardous sequences; a benchtop or do-it-yourself (DIY) builder who follows open protocols places no order, so nothing about that synthesis reaches the point where screening happens. Whether this gap is a real biosecurity problem is testable. I assess eight synthesis routes across technical and governance dimensions under the current regime (voluntary provider screening, no on-device mandate) and a projected regime (mandatory provider and on-device benchtop screening). Three findings stand out. The reproducible DIY synthesizers are academic instruments that produce short, low-fidelity oligonucleotides, and their limits are set by chemistry and laboratory infrastructure rather than by cost or fabrication skill. Restriction of inputs is not a durable control: of 34 inputs assessed, almost all are commodity and substitutable, the two hard-to-source items (a single-source micromirror device and a foundry-made electrode array) are not monitorable, and the defined-sequence enzymatic reagents are a capability barrier rather than a purchasable input, controllable only in the one commercial form where a closed benchtop vendor-locks them to a validated device. The larger exposure is the mature benchtop layer, where a United States mandate would reach only 8 of 34 known manufacturers while the newest and highest-throughput capacity concentrates outside its jurisdiction. Because prevention is necessarily incomplete, I develop a complementary capability: inferring synthesis process from the error signature of sequenced product. Reprocessing four public deposits from raw reads reproduces their published per-process signatures, and a co-processed reference atlas of 65 runs separates four synthesis processes at full accuracy under leave-one-run-out validation. Because three of the four processes are currently represented by a single co-processed deposit each, I report this four-way result as provisional, corroborate the column and photolithographic classes with independent held-out deposits, and rest the operational claim on a calibrated exclusion contrast that reliably rules out a commercial-column origin. The method degrades under post-synthesis assembly, so its scope is unassembled oligo pools. Taken together, the results argue for a layered architecture that pairs mandated screening and international harmonisation with post-hoc attribution.

Keywords: DNA synthesis screening; biosecurity; benchtop synthesizers; DIY biology; synthesis-route attribution; sequence forensics.

1. Introduction

A determined actor who wants a hazardous nucleic acid has, in principle, two routes. They can order it from a provider, where sequence screening can flag a sequence of concern (SOC) before it ships, or they can make it themselves. The screening perimeter covers the first route and not the second. Open protocols and open-hardware designs for oligonucleotide synthesis are public, benchtop instruments are sold with limited oversight, and neither is governed the way mail-order providers are. This asymmetry motivates the present assessment.

The policy setting is in transition. The 2024 Framework for Nucleic Acid Synthesis Screening from the United States Office of Science and Technology Policy took first effect for federally funded purchases in April 2025 at a 200-nucleotide window, with a tightening to 50 nucleotides and a functional SOC definition scheduled for October 2026 [1]. Executive Order 14292 then directed the framework to be revised or replaced [2], and the revision is unpublished. The International Gene Synthesis Consortium protocol remains voluntary [3], and legislation that would mandate provider screening through the Department of Commerce (S. 3741) has been introduced but not enacted [4]. I therefore analyse two regimes: the observed status quo, and a projected regime in which provider and on-device benchtop screening are mandatory. The projected regime is doubly conditional, on the framework revision issuing substantially as scheduled and on enabling legislation being enacted, and I treat it as a direction of travel rather than a fact.

The object of interest is the residual gap: the synthesis capability that remains outside the perimeter once commercial providers and benchtop manufacturers are covered. This work characterises that gap and, because a resilient system cannot rely on prevention alone [5], develops a post-hoc attribution method as the complementary detection layer. A dual-use and biosecurity statement is given before the references.

2. Synthesis routes and their readiness

Eight routes span the market from open hardware to mature commercial instruments (Table 1). Cost figures below should be read against the actor: a lone individual, a funded group, and a state programme face very different budgets, so a price that rules one out can be trivial for another. The two reproducible open synthesizers are academic instruments with limited output. OpenIDS, an open-source three-dimensional-printed inkjet system, has demonstrated oligonucleotides of roughly 15 to 30 nucleotides, with a second-generation build reduced to about $4,000 (from roughly $19,900 for the first-generation instrument) [6, 7]. Its developer attributes the short-oligonucleotide ceiling to the inkjet architecture: reagent delivered as picolitre droplets has a high surface-area-to-volume ratio and is acutely sensitive to atmospheric moisture, which is hard to exclude in a DIY setting. On his account a build using conventional column phosphoramidite chemistry would face an easier moisture problem and could plausibly reach about 100 nucleotides; this is a capability estimate, not a demonstrated result, no such open instrument has been published, and the practical barrier is the laboratory infrastructure for handling corrosive acids, organic solvents, and pressurised gases rather than the build itself (J. Kim, personal communication, 2026).

The open photolithographic platform MAS 2.0 is a maskless array synthesizer that produces library-grade material [8]; a developer interview put a working installation at EUR 200,000 to 300,000, with the optics alone near EUR 150,000, and no independent build exists outside the originating laboratory (M. Somoza, personal communication, 2026); at EUR 200,000 to 300,000 the platform is out of reach for an individual but affordable to a well-funded group or a state programme. Electrochemical synthesis rests on a single data-storage proof-of-concept at low readiness with no independent build and no commercial benchtop instance [9]. Enzymatic synthesis using terminal deoxynucleotidyl transferase avoids acids and pressurised gases and is the route most plausibly compatible with a home setting in the longer term [10], but two very different maturity levels must be kept apart. Defined-sequence enzymatic synthesis is already commercially mature in a closed benchtop form: DNA Script's SYNTAX writes up to 120 nucleotides commercially, with 500 to 1000 nucleotides demonstrated but deliberately not commercialised, at roughly 0.1% error (one in a thousand) and 384 oligonucleotides in parallel, for about $250,000 (T. Ybert, personal communication, 2026). Crucially it is a closed system: its reagents run only on the vendor's device, operation depends on cloud authorisation, and reagent supply is tied to validated customers, so a resold or diverted unit is inert, and the vendor buys instruments back rather than let them be resold. *DIY* enzymatic work, by contrast, remains limited to data storage, where it produces stochastic homopolymers rather than defined sequences [11]; a from-scratch build is gated by three reagents that must each be manufactured at high purity, the engineered enzyme, the modified nucleotides, and the solid support, which the commercial developer spent years of process chemistry to master and likens to building a graphics processor in a home kitchen, and does not regard as a credible garage threat (T. Ybert, personal communication, 2026). The same developer reframes the commercial capability itself: because building to gene length by cyclic single-strand synthesis is slow, even a 500-to-1000-nucleotide instrument does not shortcut the threat model, since the accessible route to a gene stays parallel oligonucleotide synthesis followed by routine assembly rather than de novo long synthesis on one device.

DropSynth is inexpensive gene assembly rather than de novo synthesis: it stitches microarray-derived oligonucleotides into genes and therefore consumes a commercial oligonucleotide pool [12, 13]. Its co-inventor is direct about its biosecurity relevance. The method is a massively multiplexed library tool, needs about 70 nucleotides of every input oligonucleotide for its own barcodes and overlaps, and is, in his words, "terrible" for building one specific sequence; any standard assembly method is better for a single target (C. Plesa, personal communication, 2026). Only three vendors (Twist, Agilent, Dynegene) reliably clear the roughly one-error-in-500-bases fidelity that DropSynth input requires, so the workflow is pushed onto exactly the centralised providers a mandate would reach.

3. Fidelity and the pipeline after synthesis

Assembling short oligonucleotides into genes is routine, including from oligomers only a few tens of bases long [14]. The constraint is fidelity. Standard practice targets above 90% full-length building blocks, and below about 70% a redesign is indicated, so DIY output near 56% full-length falls well short of what assembly needs [7]. Errors compound through assembly: even with good oligonucleotides, polymerase cycling assembly of a one-kilobase fragment yields only a few percent correct product before error correction, about 4% in a recent optimisation study, rising several-fold after two correction rounds [14, 15]. Recovering a correct func-

Table 1: Eight synthesis routes under assessment. Capital and length are order-of-magnitude working values; "reachable" indicates whether a provider or device mandate touches the route. Readiness: Mature = production-qualified; Demonstrated = working prototype or open build; Proof-of-concept = single lab demonstration; Not demonstrated = no defined-sequence capability shown.

Route	Readiness	Capital	Usable length	Screening reachability
OpenIDS (inkjet, DIY)	Demonstrated	about $4,000 (v2)	15–30 nt (inkjet-limited)	Only via purchased reagents/oligos
MAS 2.0 (photolithographic, DIY)	Demonstrated	EUR 200–300k	library-grade	Only via purchased reagents; supply signature
Electrochemical	Proof-of-concept	about $10–12k (est.)	13–17 nt (demo)	Not device-reachable; custom array not monitorable
Enzymatic (DIY route)	Not demonstrated	low	data-storage only	Off-the-shelf reagents; not monitorable
DropSynth (assembly)	Demonstrated	about $3,400 pool	1–3 kb assembled	Inherits provider screening via oligo input
Commercial benchtop (column)	Mature	about $60k (used $15–30k)	100–150+ nt	On-device screening under a mandate; resold units escape
Commercial benchtop (enzymatic)	Mature (closed)	about $250k	120 nt (500–1000 capable)	Closed system: reagents vendor-locked to a validated device + cloud authorisation
Mail-order provider	Mature	none	genes to genomes	The screened channel; voluntary now

tional gene from DIY-grade material therefore requires mismatch cleavage, clonal selection, and sequence verification, none of which is part of a synthesizer, and above roughly 7 to 10 kilobases reliable assembly requires host systems that many viral sequences are toxic to. These steps, not the synthesis reaction, are where the difficulty sits, which is why this assessment is scoped to synthesis and treats assembly and pathogen rescue as downstream.

4. Input controls do not close the gap

A natural governance instinct is to restrict inputs. Across all routes, almost no input functions as a durable control point. Of the 34 distinct inputs in the supply-chain inventory (Appendix A), the large majority are commodity items from many suppliers, substitutable, and subject to legitimate demand that overwhelms any synthesis-specific signal. Phosphoramidites are supplied by dozens of firms and can be produced on demand from more stable precursors [16]; acetonitrile is replaceable in DIY synthesis by propylene carbonate, an unregulated food additive [6]. The natural comparison is to explosive and narcotic precursors, which are controlled at the point of sale through licensing and purchase thresholds, but that model mostly fails here: these reagents come from many suppliers, substitute readily, and carry legitimate demand that dwarfs any synthesis-specific signal, so there is no narrow chokepoint to license, with the single exception of the enzymatic reagents noted below. Only two inputs are genuinely hard to obtain, and both are hardware: the custom complementary metal-oxide-semiconductor electrode array an electrochemical system would need, and the digital micromirror device used in photolithography, which has a single manufacturer but is sold openly. Neither can serve as a monitoring point, because their real markets are in semiconductors and projectors. A third kind of barrier is different again: the engineered enzyme and bespoke modified nucleotides that a defined-sequence enzymatic route requires are not a purchasable commodity but a co-developed capability, which makes them a capability barrier rather than a supply chokepoint, and the one input where a narrow reagent-level control could plausibly bite. In the single commercial instance of defined-sequence enzymatic synthesis this control is already operational rather than hypothetical: the closed benchtop couples proprietary reagents to a specific validated device and to cloud authorisation, so supply is bound to known customers and the reagent, not the hardware, is the lever (T. Ybert, personal communication, 2026). That is the exception that proves the rule, since it depends on a vendor-locked chemistry that commodity phosphoramidite does not have. Supply-chain restriction therefore does not provide a durable control under either regime, a conclusion that survives worst-case testing.

5. The benchtop layer and jurisdictional reach

If the garage is not the main concern, the mature benchtop layer is. Column benchtop instruments reach 100 to 150 nucleotides and beyond at high fidelity, which makes their output directly usable for assembly, and they sell for about $60,000 new and $15,000 to $30,000 second-hand [17, 18]. The most consequential result concerns jurisdiction. Of the 34 benchtop manufacturers in a separate device inventory (Appendix B), a count distinct from the input inventory above, 8 are headquartered in the United States and directly bound by a United States mandate, 12 sit in allied jurisdictions reachable only through harmonisation, and 14 fall outside both. Five of the nine firms founded since 2019 lie outside United States jurisdiction, and they include the highest-throughput, array-based class. About 35% of the inventory predates 2010, a legacy installed base that cannot be retrofitted with know-your-customer checks [19]. A secondary market compounds the problem: working instruments are listed openly with no buyer verification, and an instrument sold second-hand carries none of the controls attached to its original sale. A United States mandate consequently reaches fewer than a quarter of known manufacturers while the newest and most capable capacity concentrates where it does not, which places substantial weight on international harmonisation.

6. Screening performance and the functional frontier

Screening tools perform well against the sequences they are designed to catch. An inter-tool analysis against a blinded reference dataset found sensitivity above 95% and accuracy above 97%, with most tools screening to 50 nucleotides [20]. The demonstrated vulnerability is of a different kind: protein variants designed with generative models can retain predicted function while diverging enough in sequence to evade similarity-based screening, and after coordinated disclosure and patching a few percent of the more probably functional variants continued to escape detection [21]. This is the gap that the functional SOC provision scheduled for late 2026 is meant to address, and it is why functional detection, as distinct from sequence similarity alone, is the frontier problem.

7. Attribution from the error phenotype

Since prevention is incomplete by design, a resilient system needs a way to identify a synthesis route after the fact [5, 22]. Genetic-engineering attribution already infers the designer of a construct from design choices such as codon usage and backbone [23, 24, 25, 26], and its most successful output has been exclusion rather than identification [26]. Error profiles have separately been used to tell synthesis processes apart for quality control: a digital twin for DNA data storage distinguishes electrochemical from material-deposition synthesis by their error signatures [31]. What has not been done is to turn that observation into a forensic capability. The contribution here is a multi-process, co-processed reference atlas, leakage-aware validation, and a calibrated likelihood-ratio output framed for exclusion, rather than the bare observation that processes differ in error.

7.1 What the method is for, and where it stops

The use case is deliberately narrow. Attribution is a detection and deterrence layer in a delay-detect-defend architecture, not a substitute for screening. Its operational value is investigative triage: given an anomalous or hazardous sample, whether interdicted in transit, picked up in environmental or wastewater surveillance, or seized, an exclusion result tells an investigator where to spend effort, subpoenaing a provider's order records if a commercial origin cannot be excluded, or looking for equipment and reagents if it can. A secondary effect is deterrence through traceability, since a DIY product carries a more distinctive signature than a screened commercial one, which raises the cost of choosing the unscreened route.

The boundary matters as much as the capability. The signature lives at the oligonucleotide level and degrades as the product is processed: error correction, size selection, assembly, and clonal selection progressively remove the signature-bearing molecules, so the likelihood ratio decays toward uninformative as assembly proceeds and is effectively gone once material has been rescued into a virus or engineered into a working strain. The method therefore attributes unassembled or intercepted pools, not a finished agent. That is a real limit rather than a caveat to be minimised, and it fixes where the method belongs: at the synthesis and pre-assembly stage, complementing a screening perimeter that cannot reach every route.

7.2 Premise and method

Synthesis chemistry leaves a characteristic error signature. A single chemically synthesised batch measured across three high-fidelity polymerases gives the same error rate regardless of the readout enzyme, so the phenotype is a property of the process rather than the sequencing [27]. Sequencing error is separated from synthesis error by unique-molecular-identifier or overlap consensus [28, 29]. I reprocessed four public deposits from raw reads through one pipeline and combined them into an atlas spanning four synthesis processes: column phosphoramidite (two independent deposits, Masaki and Filges), photolithographic array (Lietard), electrochemical-deprotection array (one arm of Gimpel), and material-deposition array (the other arm of Gimpel). The electrochemical signature here comes from a commercial electrochemical-deprotection array and is distinct from the benchtop electrochemical proof-of-concept discussed in Section 2. Reprocessing reproduced the published per-process signatures to within about 20%: column phosphoramidite is deletion-dominated with a capping-driven G→A substitution bias [27]; photolithographic synthesis shows a G→T bias and a spatial gradient across the array [30]; electrochemical synthesis is the most deletion-prone, roughly an order of magnitude above material deposition [31]. Separately, within the column class the four-vendor differences are measurable but small [28], and are treated below as a within-class analysis, not as one of the four processes. A complementary attribution modality exists outside the sequence domain: because enzymatic and phosphoramidite chemistries build DNA by different mechanisms, their products differ in branching and impurity profiles that mass spectrometry can resolve, so a physical sample can in principle be assigned to a chemistry class even when only the molecule, and not its sequencing reads, is available (T. Ybert, personal communication, 2026). This physical channel is orthogonal to the read-based method developed here, separating chemistry class rather than synthesis process, and is noted as a direction to ground in the analytical-chemistry literature rather than a result demonstrated here.

7.3 Results

From these I built a single co-processed reference atlas of 65 sequencing runs (column phosphoramidite, 58 runs across two independent deposits; photolithographic, 3 runs; electrochemical array, 2 runs; material-deposition array, 2 runs; Appendix C), with shared denominators and one 16-feature extraction, verified to rebuild bit-for-bit from the per-run error tables. On this atlas the four-process classifier reaches 100% balanced accuracy under leave-one-run-out validation (chance 25%), with a label-shuffle permutation test at $p \approx 0.01$, and the processes differ by orders of magnitude in deletion rate. That headline number carries little per-class precision, and I report it with that caveat. Exact (Clopper–Pearson) 95% confidence intervals on per-class recall are tight only for the column class (58 of 58 correct, interval 0.94 to 1.00); for the three single-deposit classes, each only two or three correct calls, the intervals run as wide as 0.16 to 1.00. The four-way accuracy is therefore consistent with substantially lower true per-class accuracy, a second and statistical reason, alongside the leakage argument, to treat it as provisional.

This result must be read with an important caveat. In the co-processed atlas, three of the four processes are represented by a single deposit each, so for those classes leave-one-run-out tests run-to-run variation within one laboratory's dataset rather than generalisation across laboratories, and the classifier could in principle separate the classes on deposit- or batch-specific features, the same leakage that inflates the vendor task below. I therefore report the four-way accuracy as provisional pending at least a second independent deposit per process, and rest the paper's operational claim on the more conservative exclusion contrast rather than on positive four-way identification. The column class is the exception, and a direct test bears this out: a third independent column deposit (the Yeom SHIFT dataset) was held out entirely and passed through the same feature extraction and classifier, and all three of its rotations were classified column phosphoramidite (held-out probability of a column call between 0.49 and 0.81), a call that survives dropping the insertion feature and rests on deletion-dominance and the G→A capping bias. Because that deposit was consensus-called by a separate routine rather than the shared pipeline, it is a supportive external check rather than part of the co-processed atlas. The same was done for photolithographic synthesis: an independent second deposit from a different maskless instrument [32] was processed the same external way and, held out, was classified photolithographic (probability about 0.64). The two photolithographic deposits differ sharply in error magnitude, since that instrument is a speed-optimised synthesis carrying roughly an order of magnitude more total error than the milder Lietard deposit, so the generalisation holds from Lietard to it but not in reverse, and the class is best read as real but broad in rate. With these checks, column and photolithographic each now rest on more than one independent deposit; only the two array processes (electrochemical and material-deposition) still lack any independent replication and remain the priority for confirming the four-way result.

Cast as evidence, the binary exclusion contrast (commercial-column origin versus any other process) is well-calibrated on held-out runs, with an expected calibration error near 0.05 and complete exclusion power on the present atlas: every non-column run yields a likelihood ratio below one against a commercial-column origin (median about 48× for inclusion and 0.05× for exclusion). Bootstrapped over runs, the exclusion side is stable, with a median exclusion likelihood ratio of 0.047× (95% CI 0.041 to 0.055×), whereas the inclusion side has a wide upper tail because several column runs receive posteriors near one (median 48×, lower bound about 16×); all seven non-column runs give a likelihood ratio below one, an exclusion power whose exact 95% interval is 0.59 to 1.00 on this small non-column sample. The likelihood ratio is computed from the classifier's calibrated posterior for this binary contrast, and calibration is reported on held-out runs only; because 58 of 65 runs are column, the calibration principally reflects the column class and the non-column side rests on few runs. Within-chemistry vendor attribution is genuinely hard: under leakage-safe leave-one-lot-out validation near-neighbour vendors reach only about 72%, and an earlier run-level estimate that looked higher was inflated by replicate leakage.

7.4 Scope and limits

Several limits are structural, beyond the assembly boundary set out above. A practitioner account clarifies that assembly chemistry itself adds few errors, and that it is clonal selection that removes the signature-bearing molecules, imperfectly (C. Plesa, personal communication, 2026). DIY signatures should be a heterogeneous family rather than a single fingerprint, because builders vary reagents, protocol, and capping freely, whereas commercial providers hold conditions standardised (J. Kim, personal communication, 2026). A concrete DIY-versus-commercial discriminator, the deliberate omission of the capping step in OpenIDS, follows mechanically from the capping-driven G→A signature, but it cannot be tested on OpenIDS itself: no product reads exist and the project was discontinued, so it stands as a mechanism-anchored prediction. Finally, several load-bearing statements in Sections 2 and 7 are developer personal communications and expert estimates rather than independently verified measurements, in particular the roughly 100-nucleotide column-DIY figure, and are flagged as such where they appear. The practical output therefore remains exclusion of a standard commercial origin rather than positive DIY identification.

8. Governance implications

A single control at the point of sale is not enough, so the architecture has to be layered. Provider and on-device screening can be mandated by regulation, and their reach depends on how much of the world international harmonisation covers, which the jurisdictional finding shows is the binding variable. Whether open-source and DIY instruments fall within a device mandate is unresolved, and it is a genuine gap rather than a pending resolution: current legislation targets providers and manufacturers, not builders. For OpenIDS specifically, the complete build is public and reproducible without the developer's knowledge, so a mandate written against the design is unenforceable and the reachable handles are the wet-laboratory inputs and purchased reagents. Post-hoc attribution is the layer that acts after synthesis rather than before it, so it reaches places prevention cannot: legacy devices, the resale market, and the residual DIY routes. Physical-signature and supply-chain forensics belong in compartmented working-group and agency work; the sequence-level method developed here is the open, publishable complement.

9. Conclusion

The DIY frontier is narrower than surrounding discussion assumes. The reproducible open synthesizers are laboratory instruments producing short or library-grade output, the binding constraint sits downstream in fidelity, error correction, and host systems, and there is no evidence of a home-buildable synthesizer that works outside a laboratory. The larger exposure is the mature benchtop layer, whose output is assembly-ready, proliferating, largely unscreened, and actively resold, and whose jurisdictional position places most capacity beyond any single mandate. That exposure cannot be closed at the input. Attribution from the error phenotype, demonstrated here on real data and framed around exclusion, is the detection layer that pairs with imperfect prevention; strengthening it now depends chiefly on widening the reference atlas to independent deposits per process.

Dual-use and biosecurity statement

This work uses only published and deposited sequencing data; no nucleic acid was synthesised, and no synthesis protocol is developed or optimised. Any DIY-specific product data was gated through institutional infohazard review (IBBIS). The attribution method is itself dual-use, since an actor could in principle use it to test whether their own product is attributable, which is why the paper documents rather than develops the chemistry-level evasion route (non-canonical guanosines that locally suppress the diagnostic substitution) and frames the operational output as exclusion rather than as a synthesis recipe. No operational or procurement detail is provided.

Data and code availability

The co-processed reference atlas (65 runs, 16 features), its data dictionary and provenance, the per-run error tables it is built from, the full analysis pipeline (error calling, feature extraction, atlas build, leakage-aware classification, and likelihood-ratio calibration), and the held-out Yeom SHIFT cross-deposit check are archived at `https://doi.org/10.5281/zenodo.22014124` and mirrored at `https://github.com/olena25didenko/DIY_DNA_Synthesis`. No raw sequencing reads are redistributed; every source dataset is public and identified by accession (Filges PRJNA727098; Masaki DRA013805; Lietard PRJEB43002; Gimpel PRJEB65931), and the atlas rebuilds from the included error tables without network access. The release is subject to institutional infohazard review (IBBIS) before public posting.

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A. Full input supply-chain and regulatory-feasibility matrix

Every input a synthesis route needs, and whether it is a durable control point. “Regulate?” rates feasibility as a biosecurity control lever. Used-by abbreviations: Col = column/commercial benchtop; OID = OpenIDS inkjet; MAS = MAS 2.0 photolithographic; EC = electrochemical; EnzS/EnzB = enzymatic service/benchtop; DIY-Enz = off-the-shelf enzymatic; DS = DropSynth assembly. Of the 34 inputs, 25 are infeasible or very-low to regulate and 2 are low; 2 are access barriers that cannot be monitored (the micromirror device and the electrode array); 1 is a capability barrier rather than a purchasable input (the bespoke reversible-terminator nucleotides for defined-sequence enzymatic synthesis, row 24, with the engineered TdT in row 22 a further capability barrier in its engineered form); 3 are genuine partial chokepoints (the DropSynth oligo pool via provider screening, the proprietary cartridge, and the manufacturer, the last two eroded by the resale market); and 1 is an anti-chokepoint, the used-device aftermarket (row 34). The categories are mutually exclusive and sum to 34.

#	Compound / component	Category	Function	Used by	Sourcing & substitutability	Regulate?
1	dA/dC/dG/dT phosphoramidites (DMT-protected)	Core reagent	The four activated, protected DNA building blocks	Col, OID, MAS, EC	~45–60 suppliers worldwide (Glen Research, ChemGenes, Biosearch/LGC, Hongene, Sigma); substitutable; can be made on demand (Sandahl 2021)	Infeasible
2	Modified/RNA amidites (2′-OMe, LNA)	Core reagent	Optional modified bases	Col, MAS	Many specialty suppliers; not needed for basic DNA	Infeasible
3	Photolabile amidites (NPPOC/BzNPPOC)	Specialty reagent	5′-photocleavable protecting group	MAS	Few specialty suppliers; not commodity	Low — a rare real reagent friction, but narrow
4	Activator (1H-tetrazole/ ETT/DCI/BTT)	Cycle reagent	Activates amidite for coupling	Col, OID, MAS, EC	Standard from every oligo-reagent house; four interchangeable options	Infeasible
5	Oxidiser (iodine/ pyridine/water/THF; or CSO)	Cycle reagent	Oxidises phosphite triester to phosphate	Col, OID, MAS, EC	Iodine is a bulk commodity; non-aqueous substitute exists	Infeasible
6	Deblock acid (di-/tri-chloroacetic acid in DCM/toluene)	Cycle reagent	Removes 5′-DMT to expose next site	Col, OID, MAS	Bulk commodity chemicals	Infeasible
7	Capping A (acetic anhydride, Ac$_2$O; or Pac$_2$O)	Cycle reagent	Caps failed sequences (terminates n−1)	Col, MAS, EC	Bulk commodity; omitted in OpenIDS (forensic signature)	Infeasible
8	Capping B (N-methylimidazole, NMI)	Cycle reagent	Catalyst for capping	Col, MAS, EC	Commodity	Infeasible
9	Coupling/wash solvent (anhydrous acetonitrile)	Solvent	Reaction/wash solvent	Col, OID, MAS, EC	Substitutable by propylene carbonate (GRAS), Kim 2024	Infeasible
10	Cleavage/deprotection (aq. ammonia or AMA)	Reagent	Cleaves product; removes base protection	Col, OID, MAS, EC	Bulk commodity	Infeasible
11	Inert gas (argon/nitrogen)	Consumable	Maintains anhydrous atmosphere	Col, OID, MAS, EC	Industrial-gas commodity	Infeasible
12	Solid support (CPG/ polystyrene/universal)	Support	Anchors the growing chain	Col, OID	Multi-sourced (Prime Synthesis, Glen, ChemGenes, Kisker)	Very low
13	Functionalised glass/silica slides	Support/ substrate	Array synthesis surface	MAS	Standard microarray substrates; multi-sourced	Very low
14	Electrolyte + electrogenerated-acid system	EC reagent	Generates acid at active electrodes for deblock	EC	Standard electrochemistry reagents	Infeasible

#	Compound / component	Category	Function	Used by	Sourcing & substitutability	Regulate?
15	Industrial piezo inkjet printhead	Hardware	Deposits reagent droplets onto array spots	OID	A handful of makers (Fujifilm Dimatix, Xaar, Konica Minolta, Epson, Ricoh, Kyocera); commodity	Very low — cover demand defeats it
16	Digital micromirror device (DMD)	Hardware	Patterns UV to photo-deprotect features	MAS	Effectively single-sourced (Texas Instruments DLP)	Access barrier (not regulatable)
17	UV source ~365 nm + projection optics	Hardware	Illumination for photo-deprotection	MAS	Commodity UV LEDs (Nichia) + standard optics	Very low
18	CMOS/microelectrode array chip	Hardware	Addressable electrodes for site-selective electrochemistry	EC	Custom semiconductor fabrication (foundry + chip design)	Access barrier (not regulatable)
19	Microcontroller (Arduino/Raspberry Pi)	Hardware	Device control	OID, MAS, EC	Ubiquitous commodity electronics	Infeasible
20	Pumps/valves/PEEK manifolds/tubing/seals	Hardware	Anhydrous reagent fluidics	Col, OID, MAS, EC	Commodity lab-fluidics + webshops (OligoMaker)	Very low
21	3D-printed structural parts	Hardware	Device frame/chassis	OID, MAS	Self-produced from open CAD	Infeasible
22	Terminal deoxynucleotidyl transferase (TdT)	Enzyme	Template-independent base addition	EnzS, EnzB, DIY-Enz	Commercial (NEB, Thermo) for wild-type; engineered TdT is proprietary	Infeasible (WT); capability barrier (engineered)
23	dNTPs (natural)	Enzyme reagent	Substrate for TdT/assembly	all enzymatic, DS	Commodity	Infeasible
24	Modified/reversible-terminator dNTPs or TdT–dNTP conjugates	Specialty reagent	Enable defined-sequence enzymatic synthesis	EnzB (defined-seq)	Bespoke, co-developed with the enzyme; proprietary, multi-year	Capability barrier (not purchasable)
25	Apyrase	Enzyme	Degrades excess dNTP (kinetic control)	DIY-Enz (Lee 2019)	Commodity (Sigma)	Infeasible
26	Divalent-cation cofactor (Co^{2+}/Mg^{2+}/Mn^{2+}) + buffer	Reagent	TdT catalytic cofactor	all enzymatic	Commodity salts	Infeasible
27	Reversible-terminator cleavage reagent (e.g. TCEP)	Reagent	Unblocks 3′ end each cycle	EnzB (defined-seq)	Commodity or method-specific	Infeasible
28	Commercial microarray oligo pool	Input	The oligos DropSynth assembles into genes	DS	Ordered from a screenable provider	Partial chokepoint (via provider screening)
29	Barcoded microbeads	Consumable	Compartment barcoding for pooled assembly	DS	Specialty consumable (~$3.4K pool)	Low
30	Emulsion oil/surfactant	Consumable	Water-in-oil compartments (vortex, no microfluidics)	DS	Commodity	Infeasible

#	Compound / component	Category	Function	Used by	Sourcing & substitutability	Regulate?
31	Assembly enzymes (polymerases, T4 ligase, Type IIS)	Enzyme	PCR assembly / Golden Gate	DS	Commodity molecular-biology reagents	Infeasible
32	Proprietary reagent cartridge/kit (SYNTAX; Kilobaser chip; BioXP kit)	Consumable	Closed, licensed consumable a benchtop runs on	EnzB, some Col	Vendor-locked, licensed	Partial chokepoint (strongest benchtop lever)
33	The instrument itself (benchtop synthesizer)	Device	The synthesis engine; on-device screening point	Col, EnzB	34-firm landscape; ~8 US-bound / ~14 outside	Partial chokepoint (~8/34 reach + ~35% legacy)
34	Used/resold instruments + spare parts	Aftermarket	Second-hand devices, parts & reagents	Col, EnzB	Open resale (eBay, LabX, EquipNet); OligoMaker parts	Anti-chokepoint (routes around #32, #33)

B. Benchtop manufacturer jurisdictional reach

Summary of the ERA/IBBIS benchtop-manufacturer inventory that underlies the reach finding in Section 5. A United States on-device mandate binds only manufacturers headquartered in the United States; allied jurisdictions are reachable only through international harmonisation; the remainder are outside reach entirely.

Jurisdiction / attribute	Count	Share of 34
United States (directly bound by a US mandate)	8	~24%
Allied jurisdictions (reachable via harmonisation)	12	~35%
Outside both (beyond reach)	14	~41%
Firms founded since 2019	9	—
of which outside US jurisdiction	5	—
Inventory predating 2010 (legacy, non-retrofittable)	~12	~35%
Reach of a US-only mandate	**8 / 34**	**<25%**

Two device-side levers exist in principle (the proprietary reagent cartridge and the manufacturer, Appendix A rows 32–33), but both are first-sale levers eroded by the used-device and spare-parts aftermarket (row 34), and the manufacturer lever is jurisdictionally capped at 8 of 34.

C. Attribution reference atlas: datasets and reproduced signatures

Four public deposits were reprocessed from raw reads through a single pipeline and combined into one co-processed atlas of 65 sequencing runs with shared denominators and a common 16-feature extraction. Published per-process signatures reproduced to within about 20%. The four-process classifier reached 100% balanced accuracy under leave-one-run-out validation (chance 25%, permutation $p \approx 0.01$); this is reported as provisional because three of the four processes are represented by a single deposit each (see Section 7.2), so run-to-run separation cannot be distinguished from deposit-level batch effects for those classes. Within-column vendor attribution reached only about 72% under leave-one-lot-out and is a within-class limitation, not one of the four processes.

Process class	Characteristic signature	Runs	Source deposit
Column phosphoramidite	Deletion-dominated; G→A bias driven by capping	34	Masaki et al. 2022 (DDBJ DRA013805)
Column phosphoramidite	Within-column four-vendor differences (small)	24	Filges et al. 2021 (SRA PRJNA727098)
Photolithographic array	G→T bias; spatial gradient across array	3	Lietard et al. 2021 (ENA PRJEB43002)
Electrochemical array	Most deletion-prone; ~order of magnitude above deposition	2	Gimpel et al. 2023 (ENA PRJEB65931)
Material-deposition array	Baseline deposition error profile	2	Gimpel et al. 2023 (ENA PRJEB65931)
Total	4 processes; column has two independent deposits	**65**	

Note. The two array processes contribute only two runs each and each non-column process comes from a single deposit; cross-process separation therefore rests on order-of-magnitude deletion-rate gaps rather than on sample size or cross-laboratory replication, and widening these arms with independent deposits is the priority for confirming the four-way result. As an external cross-deposit check, a third column deposit (the Yeom SHIFT dataset) was held out and classified column phosphoramidite through the same feature pipeline; because it was consensus-called separately it is reported in Section 7.2 rather than added to these co-processed counts. The operational claim in the main text is the exclusion contrast (commercial-column origin versus not), not positive four-way identification.